

Syntax

- Terms refer to objects in a domain
- Predicates are used to make truth statements about objects

Functions

- $\mathcal{F}$ is the set of all functions.
- A function f of arity n map n terms to one term
- Constants are functions of arity 0

Terms

- functions $f \in \mathcal{F}$ refer to objects implicitly through other terms
- constants $c \in \mathcal{F}$ refer to individual objects
- variables $v \in \mathcal{V}$ are placeholders for objects

The set of terms $\mathcal{T}$ is defined inductively:

- $\forall v \in \mathcal{V} : v \in \mathcal{T}$
- $\forall c \in \mathcal{F} : c \in \mathcal{T}$
- $\forall f$ (of arity n) $\in \mathcal{F} : \forall t_1, \dots, t_n \in \mathcal{T} : f(t_1, \dots, t_n) \in \mathcal{T}$

Predicates

- $\mathcal{P}$ refers to the set of predicate symbols
- A predicate symbol $P \in \mathcal{P}$ of arity n maps n terms to an atom.
- A predicate symbol $P \in \mathcal{P}$ of arity 0 acts like a propositional variable.

Formulas

- The set of formulas is inductively defined as:
- If φ is a formula, then $\neg\varphi$ is a formula
- If φ, ψ are formulas, then $\varphi \odot \psi$ is a formula where $\odot \in \{\vee, \wedge, \rightarrow\}$
- If φ is a formula, then $\forall v \in \mathcal{V} : \mathcal{Q}x : \psi$ is a formula where $\mathcal{Q} \in \{\exists, \forall\}$

Ground Terms/Formulas

A term/formula is ground if it contains no variable.

Subformulas

- If φ is an atomic, then

$$\text{subformulas}(\varphi) = \{\varphi\}$$
- If φ is of the form $\neg\psi, \forall x : \psi, \exists x : \psi$, then

$$\text{subformulas}(\varphi) = \{\varphi\} \cup \text{subformulas}(\psi)$$
- If φ is of the from $\psi_1 \odot \psi_2$ where $\odot \in \{\vee, \wedge, \rightarrow\}$ then

$$\text{subformulas}(\varphi) = \{\varphi\} \cup \text{subformulas}(\psi_1) \cup \text{subformulas}(\psi_2)$$

A formula $\psi \in \text{subformulas}(\varphi)$ is called a proper subformula of φ , if $\varphi \neq \psi$.

Binding Strength (to avoid parentheses)

1. $\forall, \exists, \neg$, 2. $\wedge$, 3. $\vee$, 4. $\rightarrow$

Example:

$\neg A \wedge \exists x B \rightarrow C \vee D \leftrightarrow E$
is read as: $((\neg A) \wedge (\exists x B)) \rightarrow (C \vee D) \leftrightarrow E$

Quantifier scope and bound variables

Given a formula $\mathcal{Q}x : \varphi$ where $\mathcal{Q} \in \{\exists, \forall\}$:
variable $x \in \mathcal{V}$ is bound in φ
 ψ is the scope of x

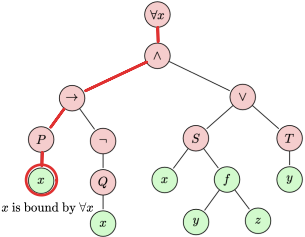
The quantifier only binds the first occurrence of a variable if outside a parentheses.

Example:

In the formula $\exists x P(x) \wedge R(x)$ only the first occurrence of x is bound, the second occurrence is free

Given a formula tree of a formula, a variable x is bound if it has an upward path to $\mathcal{Q}x$.

$\forall x((P(x) \rightarrow \neg Q(x)) \wedge (S(x, f(y, z)) \vee T(y)))$



Free variables

Given a term t , the set $\text{free}(t)$ of its free variables is defined as:

- If t is a variable $x \in \mathcal{V}$, then

$$\text{free}(t) = \{x\}$$
- If t is a constant, then

$$\text{free}(t) = \emptyset$$
- If t is of the form $f(t_1, \dots, t_n)$, then

$$\text{free}(t) = \bigcup_{i=1}^n \text{free}(t_i)$$

Given a formula φ , the set $\text{free}(\varphi)$ of its free variables is defined as:

- If φ is an atom $P(t_1, \dots, t_n)$, then

$$\text{free}(\varphi) = \bigcup_{i=1}^n \text{free}(t_i)$$
- If $\varphi = \neg\psi$, then

$$\text{free}(\varphi) = \text{free}(\psi)$$
- If $\varphi = \psi_1 \odot \psi_2$ where $\odot \in \{\vee, \wedge, \rightarrow\}$, then

$$\text{free}(\varphi) = \text{free}(\psi_1) \cup \text{free}(\psi_2)$$
- If $\varphi = \mathcal{Q}x : \psi$ where $\mathcal{Q} \in \{\forall, \exists\}$, then

$$\text{free}(\varphi) = \text{free}(\psi) \setminus \{x\}$$

Sentence

A formula is called a sentence if it contains no free variables.

Theory

A set $\Gamma = \{\varphi_1, \varphi_2, \dots\}$ of sentences is called a theory

Structures

A structure $\mathcal{M} = (\mathcal{D}, \mathcal{I})$ is made up of

- a domain $\mathcal{D}$
- an interpretation function $\mathcal{I}$

Constants, functions and predicates define the language (also called signature) $\mathcal{L}$ of the structure.

Domains

A domain $\mathcal{D}$ is a non-empty set of objects

Interpretation Function

- The interpretation function $\mathcal{I}$ maps:
- each constant c to an object $\mathcal{I}(c) = D$
- each function f of arity n to a total function $\mathcal{I}(f) = D^n \rightarrow D$
- each predicate symbol P of arity n to a relation $\mathcal{I}(P) \subseteq D^n$

One can also write $c^{\mathcal{M}}, f^{\mathcal{M}}, P^{\mathcal{M}}$ instead of $\mathcal{I}(\dots)$.

Satisfaction of ground formulas

For a structure $\mathcal{M} = (\mathcal{D}, \mathcal{I})$ and an atom $P(t_1, \dots, t_n)$ it holds:

$$\mathcal{M} \models P(t_1, \dots, t_n) \iff (t_1^{\mathcal{M}}, \dots, t_n^{\mathcal{M}}) \in P^{\mathcal{M}}$$

Furthermore, for any ground formula φ it holds:

- $\mathcal{M} \models \neg\varphi \iff \neg(\mathcal{M} \models \varphi)$
- $\mathcal{M} \models \varphi \odot \psi \iff (\mathcal{M} \models \varphi) \odot (\mathcal{M} \models \psi)$ for $\odot \in \{\vee, \wedge, \rightarrow\}$

Given only a structure, one can only prove for ground formulas

Satisfaction of Formulas

Given an interpretation $(\mathcal{M}, \ell)$ the satisfaction of formulas is defined as:

- $(\mathcal{M}, \ell) \models P(t_1, \dots, t_n) \iff (t_1^{(\mathcal{M}, \ell)}, \dots, t_n^{(\mathcal{M}, \ell)}) \in P^{\mathcal{M}}$
- $(\mathcal{M}, \ell) \models \neg\varphi \iff \neg((\mathcal{M}, \ell) \models \varphi)$
- $(\mathcal{M}, \ell) \models \varphi \odot \psi \iff ((\mathcal{M}, \ell) \models \varphi) \odot ((\mathcal{M}, \ell) \models \psi)$
- $(\mathcal{M}, \ell) \models \mathcal{Q}x : \varphi \iff \mathcal{Q}x : ((\mathcal{M}, \ell) \models \varphi)$ for $\mathcal{Q} \in \{\exists, \forall\}$

One may also write:

- $(\mathcal{M}, \ell) \models \varphi \iff \neg((\mathcal{M}, \ell) \models \neg\varphi), (\mathcal{M}, \ell) \models \varphi \iff \mathcal{M} \models_{\ell} \varphi$

Satisfaction of Sentences

- Theorem**
- Given a structure $\mathcal{M}$, a formula φ and variable assignments ℓ, ℓ' , it holds:
 $(\forall x \in \text{free}(\varphi) : \ell(x) = \ell'(x)) \implies (((\mathcal{M}, \ell) \models \varphi) \iff ((\mathcal{M}, \ell') \models \varphi))$

Given a structure $\mathcal{M}$ and a sentence φ , it holds:

- $\forall \ell, \ell' : ((\mathcal{M}, \ell) \models \varphi) \iff ((\mathcal{M}, \ell') \models \varphi)$

Satisfiability Definition

A formula φ is satisfiable $\iff \exists(\mathcal{M}, \ell) : (\mathcal{M}, \ell) \models \varphi$

Consistent Theory

A theory Γ is consistent $\iff \exists \mathcal{M} : \forall \varphi \in \Gamma : \mathcal{M} \models \varphi$

- Theorem:** $(\forall \varphi \in \Gamma : \mathcal{M} \models \varphi) \iff \left(\mathcal{M} \models \bigwedge_{\varphi \in \Gamma} \varphi \right)$

Tableau

Same inference rules as PL tableau but with additional rules:

$\mathbf{t} : \forall x F$	$\mathbf{f} : \exists x F$	$\mathbf{f} : \forall x F$	$\mathbf{t} : \exists x F$
$\mathbf{t} : F(t/x)$	$\mathbf{f} : F(t/x)$	$\mathbf{f} : F(c/x)$	$\mathbf{t} : F(c/x)$

For any variable-free term $t \in \Sigma^{\text{param}} \cup \Sigma$ (usually picking a term) already in tableau branch.

For a new Parameter $c \in \Sigma^{\text{param}}$ not occurring in the tableau branch

Note that for every ground-term in the branch the rule must be applied

$F(t/x)$ denotes the formula F where all occurrences of x are replaced by t .

Undecidability of FOL (example)

Due to the $\mathbf{t} : \forall x F$ and $\mathbf{f} : \exists x F$ rules the tableau may expand forever, as can be seen with the following example. A tableau is only guaranteed to terminate if the formula is a contradiction.

Given an interpretation $(\mathcal{M}, \ell)$ and the formula: $\forall x : P(x, g(x))$, the tableau would expand as follows:

$\mathbf{t} : \forall x : P(x, g(x))$ // Applying the inference rule by replacing x for a term c
 $\mathbf{t} : P(c, g(c))$ // since $g(c) = c'$ maps to a term, a new ground term has appeared in the branch for which this rule must be applied
 $\mathbf{t} : P(c, g(c'))$ // once again, a new ground term $g(c') = c''$ has appeared in the branch. This will never stop

Example of a structure

- constants: zero
 - functions: succ (arity 1)
 - predicates: nonneg (arity 1), equals (arity 2)
- $\mathcal{D} = \{a, aa, aaa, \dots\}$
 $\text{zero}^{\mathcal{M}} = a$
 $\forall d \in \mathcal{D} : \text{succ}^{\mathcal{M}}(d) = da$
 $\text{nonneg}^{\mathcal{M}} = \{aa, aaa, \dots\}$
 $\text{equals}^{\mathcal{M}} = \{(d, d) \mid d \in \mathcal{D}\}$

The structure satisfies the following ground atom:

- $\text{equals}(aa, \text{succ}(a))$

since $\text{succ}(a) = aa$ and $(aa, aa) \in \text{equals}^{\mathcal{M}}$

FOL Interpretation

An interpretation is a pair $(\mathcal{M}, \ell)$.

An interpretation of a term $t^{(\mathcal{M}, \ell)}$ is defined as:

- $t^{(\mathcal{M}, \ell)} = \ell(t)$ if $t \in \mathcal{V}$
- $t^{(\mathcal{M}, \ell)} = f^{\mathcal{M}}(t_1^{(\mathcal{M}, \ell)}, \dots, t_n^{(\mathcal{M}, \ell)})$ if $t = f(t_1, \dots, t_n)$

Modelling natural language

- All As are B

$$\forall x : (A(x) \rightarrow B(x))$$
- Some As are Bs

$$\exists x : (A(x) \wedge B(x))$$

- No As are B

$$\neg \exists x : (A(x) \wedge B(x))$$

- Some As are not Bs

$$\exists x : (A(x) \wedge \neg B(x))$$

(Typically) adjectives and verbs are predicates; nouns are terms

Models and Consequences

Each theory Γ has:

- Models: $\text{Mods}(\Gamma) = \{\mathcal{M} \mid \mathcal{M} \text{ is a structure, } \mathcal{M} \models \Gamma\}$
- Consequences: $\text{Cons}(\Gamma) = \{\varphi \mid \varphi \text{ is a sentence, } \Gamma \models \varphi\}$

Monotonicity Theorem:

$\text{Mods}(\Gamma \cup \{\varphi\}) \subseteq \text{Mods}(\Gamma), \text{Cons}(\Gamma) \subseteq \text{Cons}(\Gamma \cup \{\varphi\})$
 $(\Gamma \models \varphi) \implies (\forall \Gamma' : (\Gamma \subseteq \Gamma') \implies (\Gamma' \models \varphi))$

Formulas φ, ψ are equivalent $\varphi \equiv \psi \iff \text{Mods}(\varphi) = \text{Mods}(\psi)$

Herbrand Interpretation

A Herbrand interpretation is an interpretation which maps functions to applications of functions on terms

A Herbrand universe is the set of all ground terms.

Example: (same idea as before)

$\mathcal{D} = \{\text{zero}, \text{succ}(\text{zero}), \text{succ}(\text{succ}(\text{zero})), \dots\}$
 $\text{zero}^{\mathcal{M}} = \text{zero}$
 $\dots$

Variable Assignment

Given a structure $\mathcal{M}$, for a variable assignment $\ell : \mathcal{V} \rightarrow \mathcal{D}$ holds:

- Given variable $x \in \mathcal{V}$ and object $d \in \mathcal{D}$:
 $\ell[d/x]$ is a variable assignment such that

$$\ell[d/x](y) = \begin{cases} d & \text{if } y = x \\ \ell(y) & \text{if } y \neq x \end{cases}$$

Modelling natural language (Examples)

- One vehicle passed all inspections:

$$\exists x : (\text{Vehicle}(x) \wedge \forall y : (\text{Inspection}(y) \rightarrow \text{Passed}(x, y)))$$
- There is one vehicle in every lane:

$$\forall x : (\text{Lane}(x) \rightarrow \exists y : (\text{Vehicle}(y) \wedge \text{InLane}(y, x)))$$
- All the children eat a sandwich

$$\forall x : (\text{Child}(x) \rightarrow \exists y : (\text{Sandwich}(y) \wedge \text{Eats}(x, y)))$$
- All the children are in the playground:

$$\exists y : (\text{Playground}(y) \wedge \forall x : (\text{Child}(x) \rightarrow \text{In}(x, y)))$$

The robot removes a block. Then it places a block on the table:

$$\exists x_1 : (\text{Block}(x_1) \wedge \text{Removes}(\text{robot}, x_1)) \wedge \exists x_2 : (\text{Block}(x_2) \wedge \text{PlacesOnTable}(\text{robot}, x_2))$$

Equivalences of quantified formulas

$\forall x : \varphi \equiv \exists x : \neg\varphi$ $\neg \exists x : \varphi \equiv \forall x : \neg\varphi$
 $\forall x : \varphi \wedge \forall x : \psi \equiv \forall x : (\varphi \wedge \psi)$ $\exists x : \varphi \vee \exists x : \psi \equiv \exists x : (\varphi \vee \psi)$
 $\mathcal{Q}x : \mathcal{Q}y : \varphi \equiv \mathcal{Q}y : \mathcal{Q}x : \varphi$

If x not free in ψ :

$\psi \odot \mathcal{Q}x : \varphi \equiv \mathcal{Q}x : (\varphi \odot \psi)$ for $\mathcal{Q} \in \{\forall, \exists\}, \odot \in \{\vee, \wedge\}$

Negation Normal Form

A formula φ is in NNF if the only connectives are $\odot \in \{\vee, \wedge, \neg\}$ where $\neg$ is only used on variables

Prenex Normal Form

A formula $\varphi = \mathcal{Q}_1 x_1 \dots \mathcal{Q}_n x_n : \psi$ is in PNF with:

- $\mathcal{Q}_i \in \{\forall, \exists\}$ for $i = 1, \dots, n$ (called quantifier prefix)
- ψ is a quantifier-free formula (called matrix)

Computing PNF

- Eliminate all occurrences of $\rightarrow$
- Move all negations inwards
- Rename all variables apart
- Move quantifiers to the front

Example of computing the PNF

Given the formula: $\forall x : (\exists y : R(x, y) \wedge \forall y : \neg S(x, y) \rightarrow \neg(\exists y : R(x, y) \wedge P))$

- Step 1:

$$\forall x : (\neg(\exists y : R(x, y) \wedge \forall y : S(x, y)) \vee \neg(\exists y : R(x, y) \wedge P))$$
- Step 2:

$$\forall x : (\forall y : \neg R(x, y) \vee \exists y : S(x, y) \vee \forall y : \neg R(x, y) \vee \neg P)$$
- Step 3:

$$\forall x : (\forall y_1 : \neg R(x, y_1) \vee \exists y_2 : S(x, y_2) \vee \forall y_3 : \neg R(x, y_3) \vee \neg P)$$
- Step 4:

$$\forall x : \forall y_1 : \exists y_2 : \forall y_3 : (\neg R(x, y_1) \vee S(x, y_2) \vee \neg R(x, y_3) \vee \neg P)$$